Predicting Sleep Quality from Daytime Wearable Metrics: An N-of-1 Machine Learning Study

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Introduction

Sleep quality is fundamental to cognitive function and metabolic health. Consumer wearables offer continuous physiological monitoring as an alternative to PSG.

Dataset: 158 consecutive nights (Nov 2025 – May 2026), Didiconn wrist-worn device.
Features (13): HR, HRV, SpO_2, steps, calories.
Task A – **Regression**: total sleep, REM, deep sleep (minutes).
Task B – **Classification**: poor sleep night (≤82.25%, data-driven 25th percentile).

Research questions:
(1) Can next-night sleep be predicted from daytime wearable data?
(2) Which physiological indicators are early predictors of poor sleep?
(3) Does low daytime HRV precede fragmented sleep?

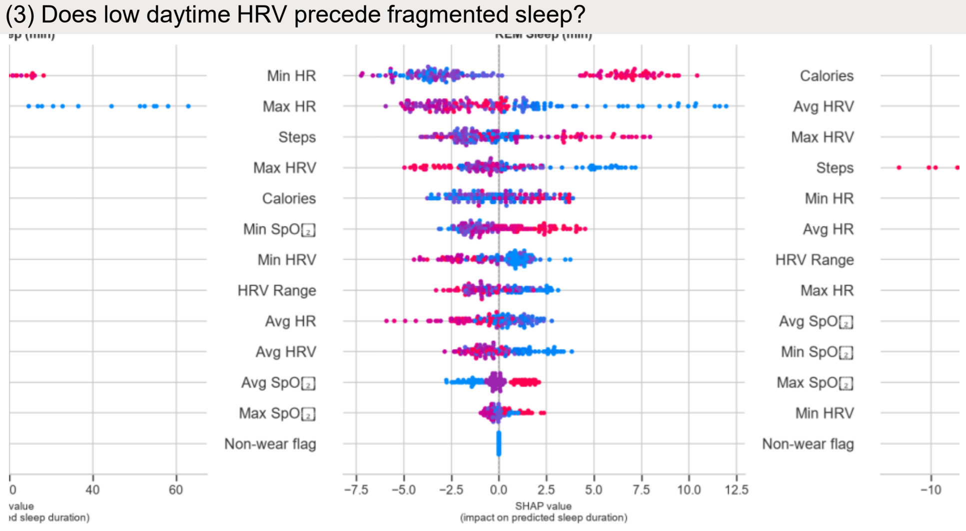

Fig. 1: SHAP beeswarm — feature importance for Total Sleep, REM, Deep Sleep. Calories and HRV dominate. Red = high feature value, blue = low.

Method

Preprocessing: t-1 lag alignment, nap deduplication, HRV range feature derived.
Validation: Walk-forward CV (TimeSeriesSplit, k=5, ~26 test nights/fold).
Random k-fold excluded — temporal leakage prevention.

ML models: Linear Regression, Ridge, Lasso, Random Forest, XGBoost (regression);
Logistic Regression, SVM (RBF), RF, XGBoost (classification). class_weight=balanced.
DL models: LSTM + 1D-CNN, 7-day sliding window (N=151 sequences), Adam lr=1e-3.

Additional experiments (6): rolling features (49-dim), SOL/WASO targets, weekly patterns (Kruskal-Wallis), nested GridSearchCV tuning, multi-output regression, anomaly detection (GMM + Isolation Forest).
XAI: TreeSHAP — global importance, HRV dependence plots, waterfall explanations.

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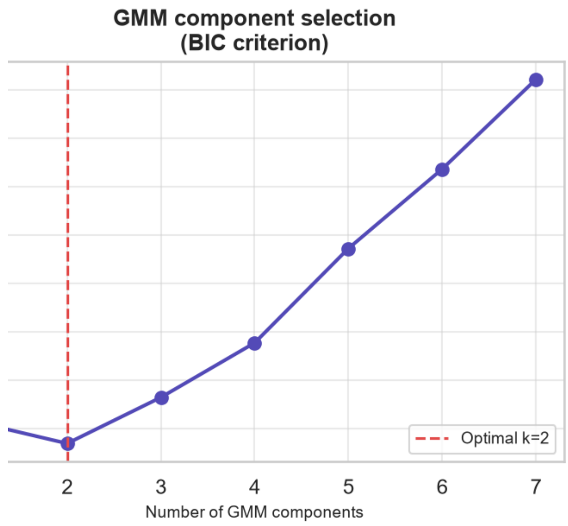

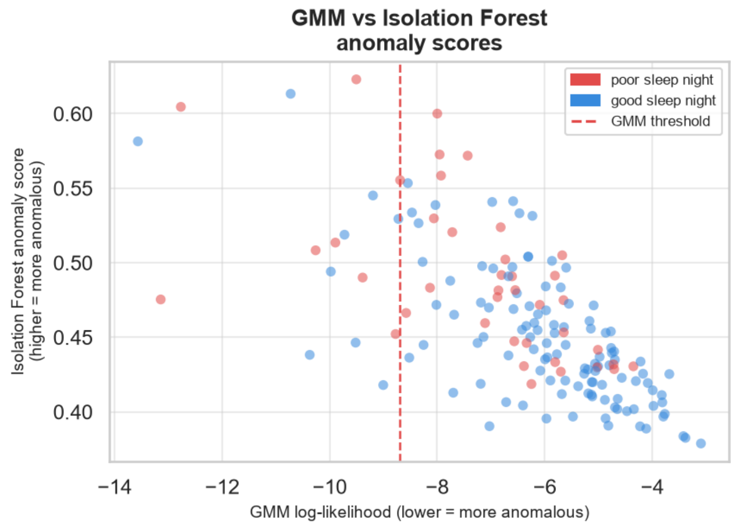

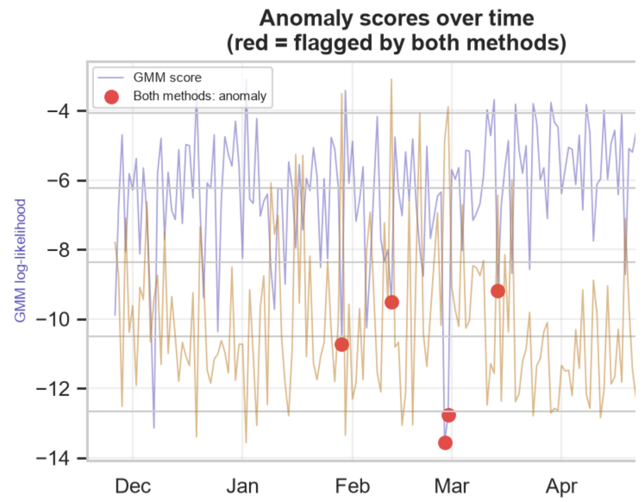

Fig. 2: Anomaly detection. Left: BIC curve (k=2 optimal). Centre: GMM vs IF scores by poor/good sleep label. Right: scores over 6 months.

Results

Regression — all models $R^2 < 0$ (worse than mean-prediction baseline):
- Random Forest: deep sleep R^2=−0.138, total sleep R^2=−0.222
- XGBoost (tuned): total sleep R^2=−0.179 (Δ+0.177 vs default)
- RF + rolling features: deep sleep R^2=−0.043 (Δ+0.095) ← largest gain

Classification — best supervised: SVM RBF AUC=0.622. F1 low (0.10–0.22).
DL underperformed: LSTM AUC=0.446 (151 sequences insufficient).

Anomaly detection outperformed ALL supervised models:
- Isolation Forest AUC=0.728 | GMM AUC=0.696
- 86.1% agreement; 5 nights flagged by both simultaneously

SHAP: Calories > Max HR > Max HRV > Steps.
HRV range = #1 feature for poor-sleep classification (|SHAP|=0.042).
Higher daytime HRV → less predicted sleep (r≈−0.75) — post-exercise confound.

Weekly: sleep efficiency significant day-of-week effect (KW H=13.94, p=0.030).

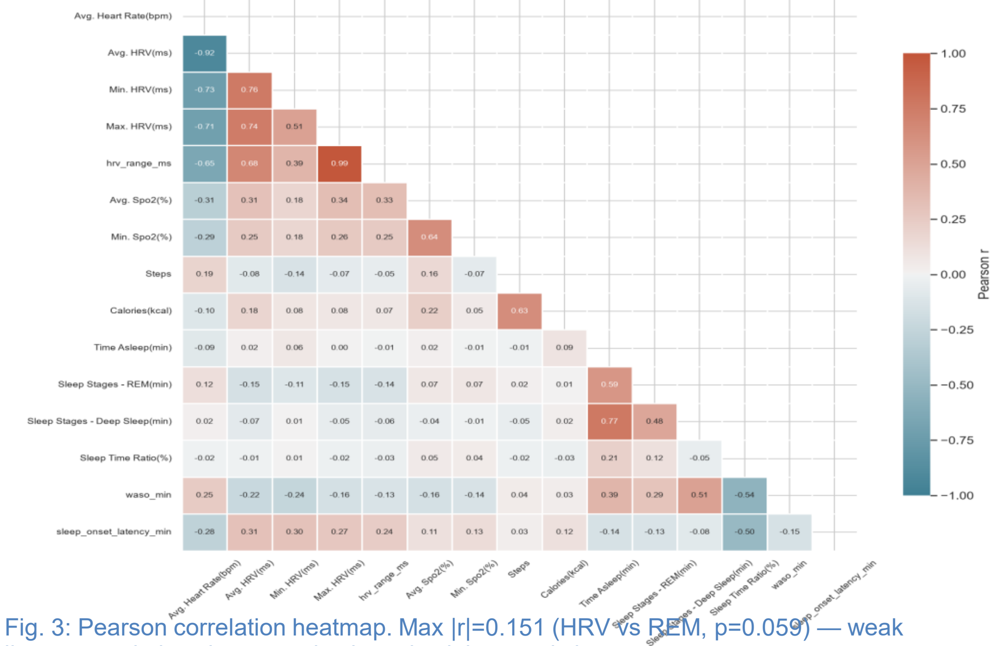

Fig. 3: Pearson correlation heatmap. Max |r|=0.151 (HRV vs REM, p=0.059) — weak linear associations between daytime physiology and sleep outcomes.

Model Performance Summary

Deep Sleep: RF+Rolling **R^2=−0.043** (best)
Total Sleep: XGBoost (tuned) **R^2=−0.179**
REM: Random Forest **R^2=−0.275**
Classification: SVM **AUC=0.622**
Anomaly: IF AUC=0.728 ← best overall

Weekly patterns: Sleep efficiency showed significant day-of-week variation (Kruskal-Wallis H=13.94, p=0.030). Wednesday consistently worst.

DL underperformed: 151 sequences insufficient for LSTM/CNN to learn temporal dependencies.

Conclusions

Single-day wearable metrics show limited predictive power for sleep (all $R^2 < 0$), consistent with the multifactorial nature of sleep regulation.

Unsupervised anomaly detection (IF AUC=0.728) outperformed all supervised classifiers (best AUC=0.622) — poor sleep nights are structurally distinguishable in architecture.

Physiological trends > single values: rolling features gave largest R^2 gain (+0.095).

HRV variability (range), not average HRV, is the strongest poor-sleep classifier feature.

Limitations: N-of-1 (single participant), device not PSG-validated, daily HRV averages conflate resting/exercise/stress.

Future work: longer observation, continuous pre-sleep HRV, ecological momentary assessment.

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